

Potentiometer

DETECT

INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses



Security-light adjustment

DETECT
Read
sensitivity
setting



DECIDE
Is movement
strong
enough?



DO
Switch the
light on



Alien voice tuner

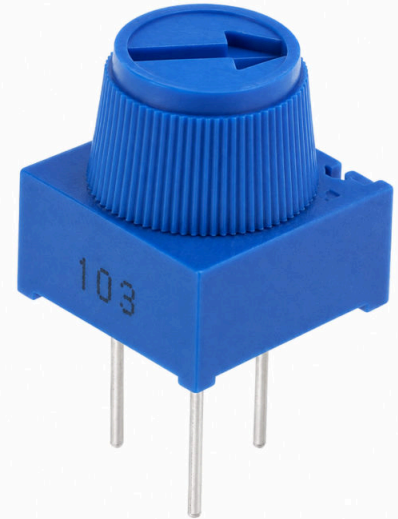
DETECT
Knob position



DECIDE
Should it
rumble or
squeak?



DO
Rumble or
squeak



Knob
Turn to adjust

Outer pair
Pins 1 and 3

Wiper
Pin 2, offset

Code

- Raspberry Pi

```
from gpiozero import MCP3008
pot = MCP3008(0, select_pin=22)
setting = pot.value
```

- **Raspberry Pi Pico**

```
from piczero import Pot

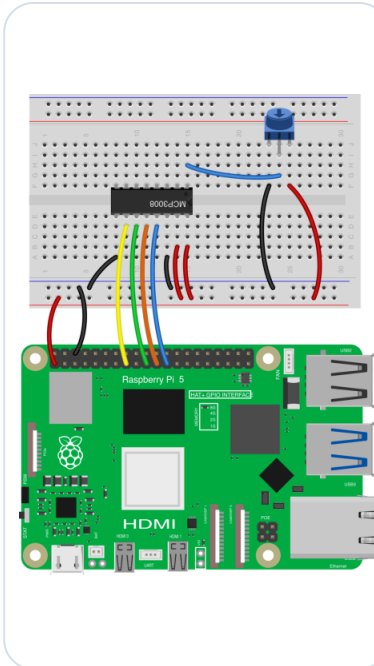
pot = Pot(26)

setting = pot.value
```

`setting` runs from 0 to 1. Read `pot.value` again after turning; `pot.voltage` gives volts. The Pi needs the MCP3008 converter shown. Power the trimpot and converter from **3V3**.

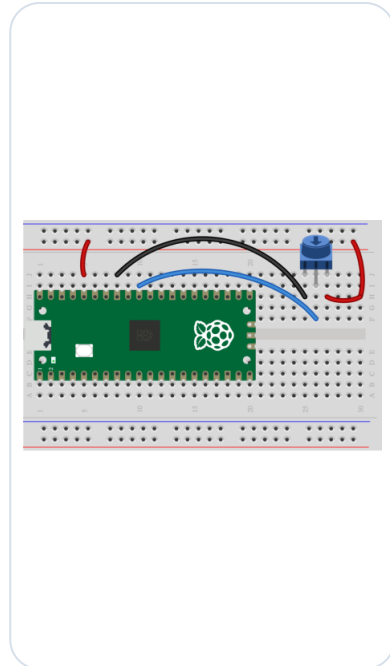
Wiring

PI 5 + MCP3008



Wiper → **CH0**. CLK / DIN / DOUT / CS → **GPIO 11 / 10 / 9 / 22**. Chip notch faces right.

PICO



Pin 1 → **GND** · 2 (wiper) → **GP26** · 3 → **3V3**. Each pin has its own breadboard row.